

Question ID: 566759ef

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Medium

Question

Thomas installed a new stove in his restaurant. At the time of installation, the stove had a value of \$800. Thomas estimates that each year the value of the stove will depreciate by 20% of the previous year’s estimated value. What is the estimated value of the stove exactly 2 years after Thomas installed it?

Answer

- A. \$480
- B. \$512
- C. \$556
- D. \$640

Rationale

Choice B is correct. If the stove’s value depreciates by 20% of the previous year’s estimated value, then each year it retains $100\% - 20\% = 80\%$, or 0.80, of the previous year’s estimated value. Since the stove’s value was \$800 when Thomas installed it, the estimated value after two years would be $(0.80)(0.80)(\$800) = \512 .

Choice A is incorrect. This is the value of the stove if each year it had depreciated by 20% of the original value rather than by 20% of the previous year’s estimated value. Choice C is incorrect and may be the result of a computational error. Choice D is incorrect. This is the estimated value of the stove 1 year after Thomas installed it, not 2 years.